

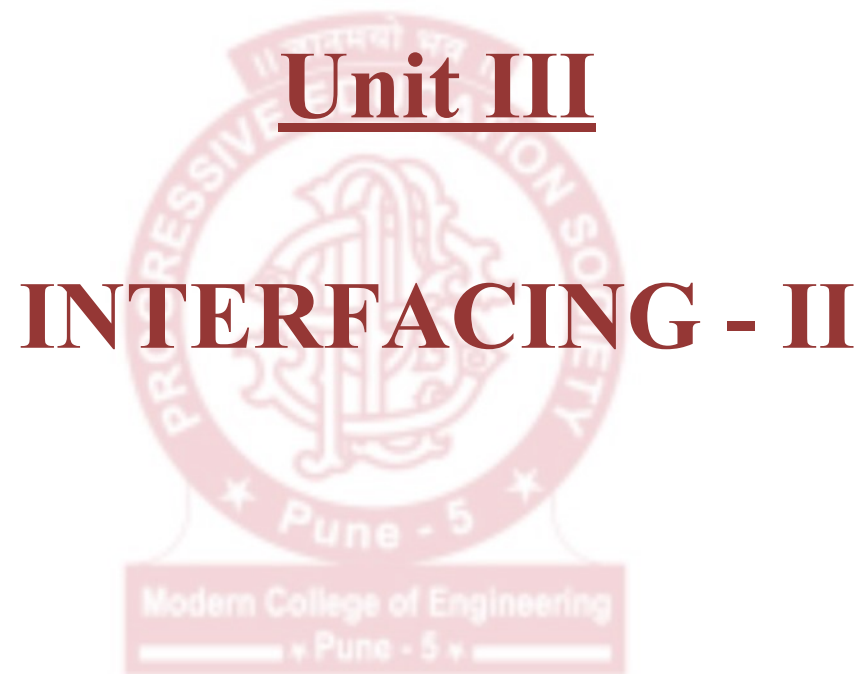


Progressive Education Society's
Modern College of Engineering, Pune - 411005
Department of Electronics & Telecommunication

AY-2024-25 Semester - I

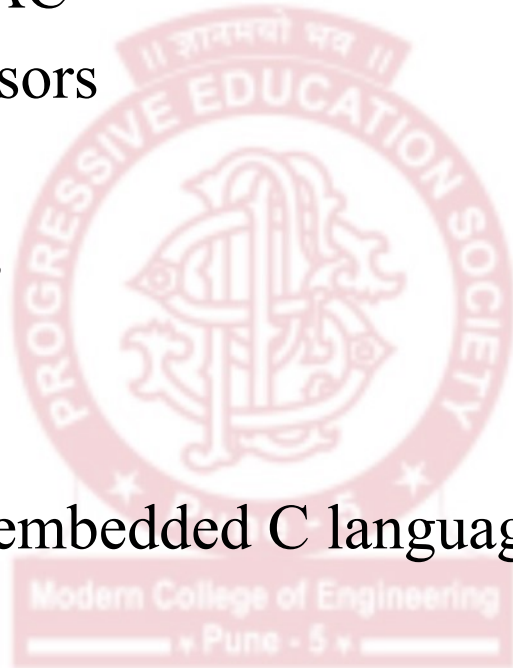
Microcontroller and Applications

Course In-charge
Mrs. R. S. Chaudhari

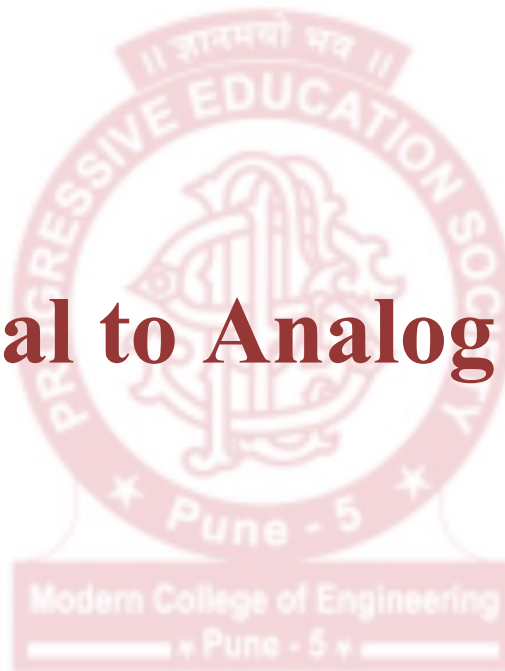


Syllabus

- Data transmission and reception using Serial port with PC
- Interfacing of DAC
- Temperature sensors
- Stepper motor
- Motion detectors
- Relay, Buzzer
- Opto-isolators
- All programs in embedded C language.

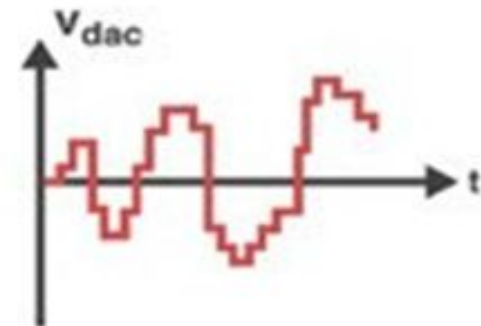
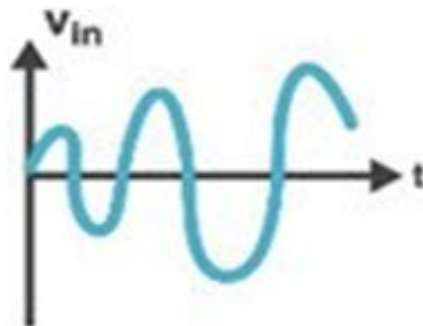


Digital to Analog Converter

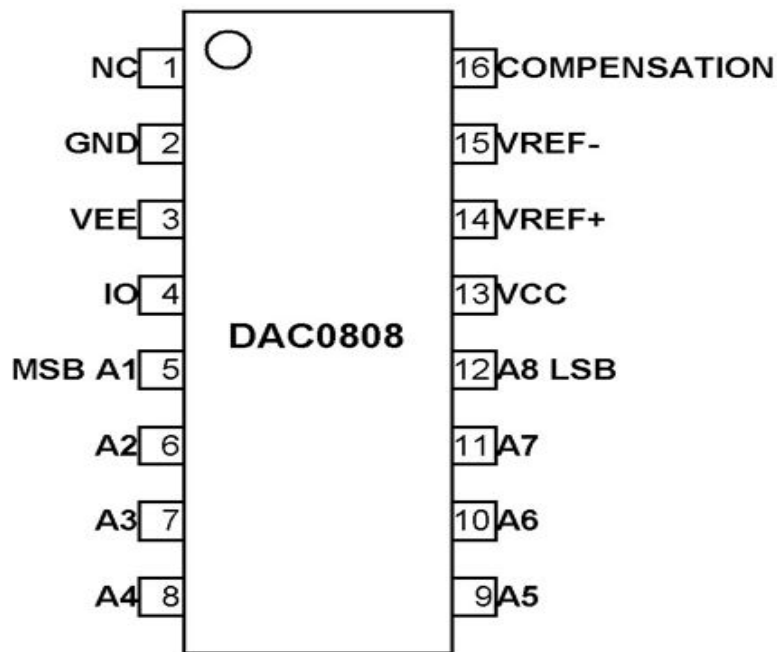


DAC and its application

Need of conversion



DAC 0808



Pin	Name	Description
1	NC	No connection
2	GND	Ground
3	VEE	Negative power supply
4	IO	Output signal pin
5	A1	Digital input bit 1 (Most Significant Bit)
6	A2	Digital input bit 2
7	A3	Digital input bit 3
8	A4	Digital input bit 4
9	A5	Digital input bit 5
10	A6	Digital input bit 6
11	A7	Digital input bit 7
12	A8	Digital input bit 8 (Least Significant Bit)
13	VCC	Positive power supply
14	VREF+	Positive reference voltage
15	VREF-	Negative reference voltage
16	COMPENSATION	Compensation capacitor pin

Features and Electrical characteristics of DAC0808

- 8 bit parallel digital data input
- Fast settling time (typical value): 150 ns
- Relative accuracy at $\pm 0.19\%$ maximum error
- Full scale current match: ± 1 LSB
- Non-inverting digital inputs are TTL and CMOS compatible
- High speed multiplying input slew rate: 8 mA/ μ s
- Power supply voltage range: ± 4.5 V to ± 18 V
- Low power consumption: 33 mW@ ± 5 V
- Maximum Power dissipation: 1000 mW
- Operating temperature range: 0°C to $+75^{\circ}\text{C}$

DAC

Types of DAC

Binary Weighted

R-2R

Output is in the form of steps

Accepts parallel data as input and provides analog current as output.

Number of bit DAC is will decide resolution for that DAC

$I_{out} =$

$I_{ref}(D_7/2 + D_6/4 + D_5/8 + D_4/16 + D_3/32 + D_2/64 + D_1/128 + D_0/256)$

Current I_{ref} is 2 mA.

Current at output is converted to input using an operational amplifier.

DAC

O/P voltage of DAC is given by:

$$V_0 = V_{ref} \left[\frac{D_0}{2} + \frac{D_1}{4} + \frac{D_2}{8} + \frac{D_3}{16} + \frac{D_4}{32} + \frac{D_5}{64} + \frac{D_6}{128} + \frac{D_7}{256} \right]$$

If Data = 00H, O/P voltage of DAC is given by: 0V

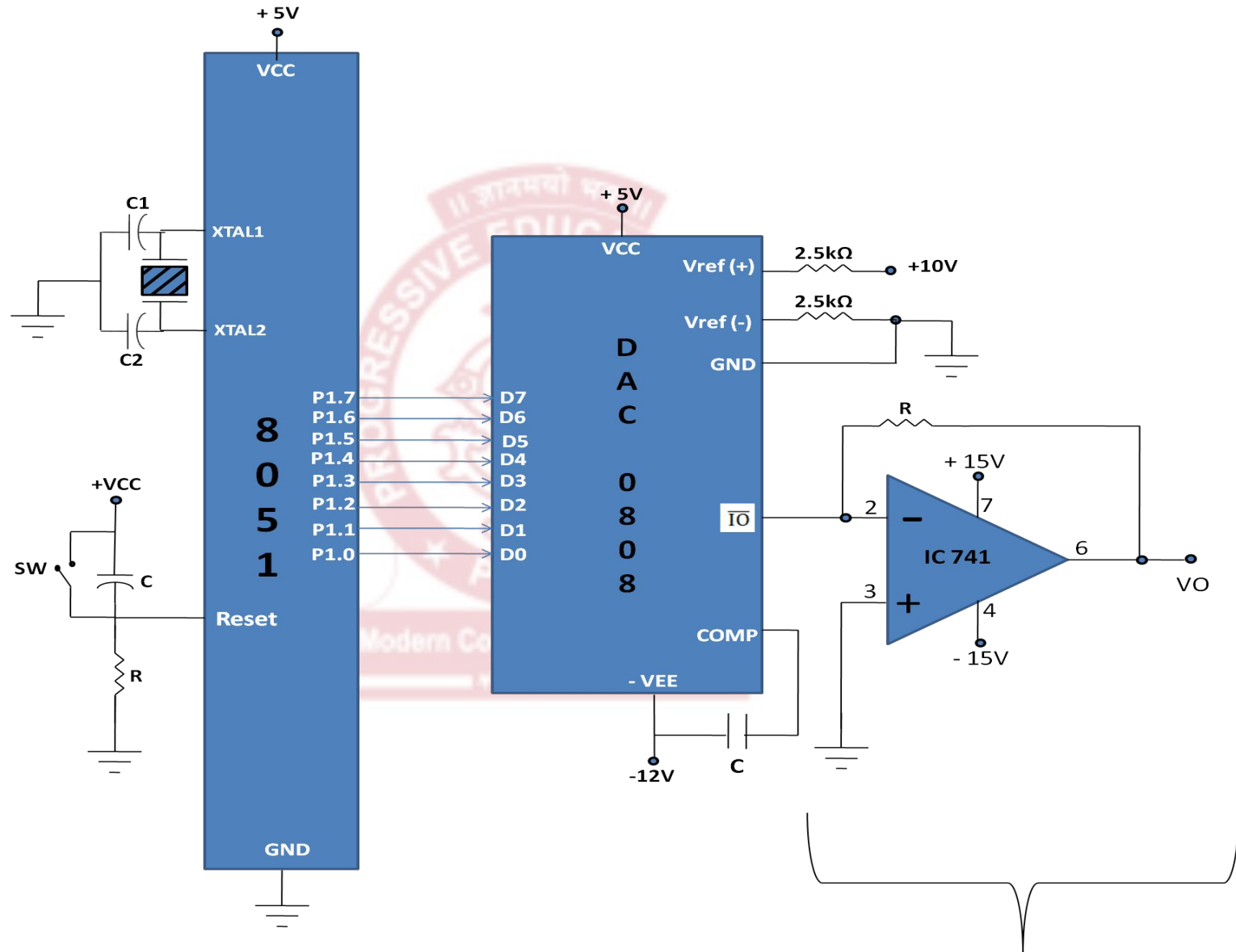
$$V_0 = V_{ref} \left[\frac{D_0}{2} + \frac{D_1}{4} + \frac{D_2}{8} + \frac{D_3}{16} + \frac{D_4}{32} + \frac{D_5}{64} + \frac{D_6}{128} + \frac{D_7}{256} \right]$$

If Data = 80H, O/P voltage of DAC is given by: 5V

$$V_0 = 10 \left[\frac{1}{2} + \frac{0}{4} + \frac{0}{8} + \frac{0}{16} + \frac{0}{32} + \frac{0}{64} + \frac{0}{128} + \frac{0}{256} \right]$$

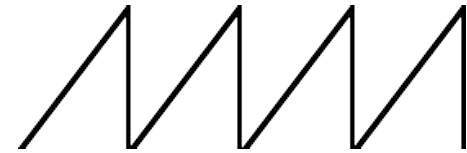
If Data = FFH, O/P voltage of DAC is given by: ?V

Interfacing of DAC with 8051



Program to generate a sawtooth waveform of 4V with 5V Vref

```
MOV R0, #XXH
REPEAT: MOV A, #00H
BACK:   MOV P1, A
        INC A
        CJNE A, R0, BACK
        SJMP REPEAT
```



Peak amplitude for sawtooth waveform is 3.906v in this case as 64 = 01100100B

$$I_{out} = I_{ref} * (0/2 + 1/4 + 1/8 + 0/16 \dots)$$

$$I_{out} = 0.781 \text{mA}$$

$$V_{out} = I_{out} * 5k = 3.906 \text{V approx} = 4 \text{Volts}$$

If our value is FF instead of 64 then peak value is 9.96V.

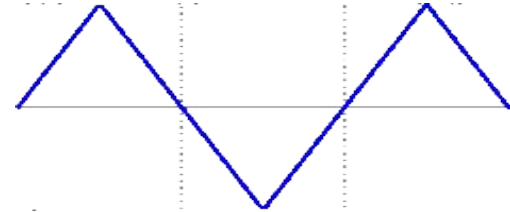
$$\text{As FF} = 11111111$$

$$I_{ou} = 1.99 \text{mA}$$

$$V_{out} = 1.99 \text{mA} * 5K = 9.96 \text{V}$$

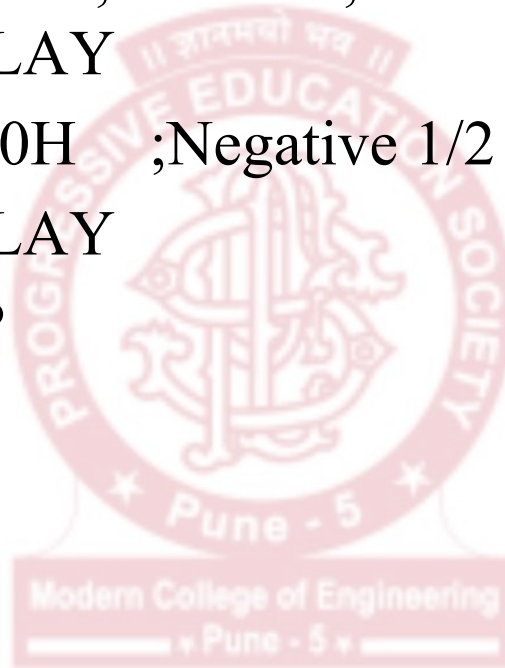
Program to generate a Triangular waveform

```
MOV A,#00H
LOOP: MOV P0,A
      INC A
      CJNE A, #0FFH,LOOP
LOOP1: MOV P0,A
      DEC A
      CJNE A, #00H, LOOP1
      SJMP LOOP
```



Program to generate a Square waveform

```
LOOP: MOV P0, #0FFH ;Positive 1/2 Cycle  
      ACALL DELAY  
      MOV P0, #00H ;Negative 1/2 Cycle  
      ACALL DELAY  
      SJMP LOOP
```



Program to generate Sine wave

- To generate a sine wave using DAC we need to know voltage magnitude at various angles from 0 to 360 degrees.
- The values of sine function vary from -1 to +1 for 0 to 360 degree.
- Thus we calculate the integer values representing the voltage magnitude for each angle theta.
- We assume that full scale output voltage for DAC is 10V
- We calculate Vout using formula $V_{out} = 5V + (5V * \sin \theta)$
- Question?? Why this formula right??

Ans: We have assumed sine function to vary from -1 to +1.

Thus at -1, $\sin(-1)$ is -0.841 and at +1 $\sin(+1)$ is 0.841

Thus, if we substitute our result is approximately 10V which is our full scale voltage.

Calculation

- For 30 degree

$$\begin{aligned}V_{out} &= 5V + (5V * \sin(30)) \\&= 5V + (5V * 0.5) \\&= 7.5 V\end{aligned}$$

For 60 degree

$$\begin{aligned}V_{out} &= 5V + (5V * \sin(60)) \\&= 5V + (5V * 0.866) \\&= 9.33V\end{aligned}$$

Values sent to DAC are decimal values

$$\text{Step size} = 256/10V = 25.60$$

Thus,

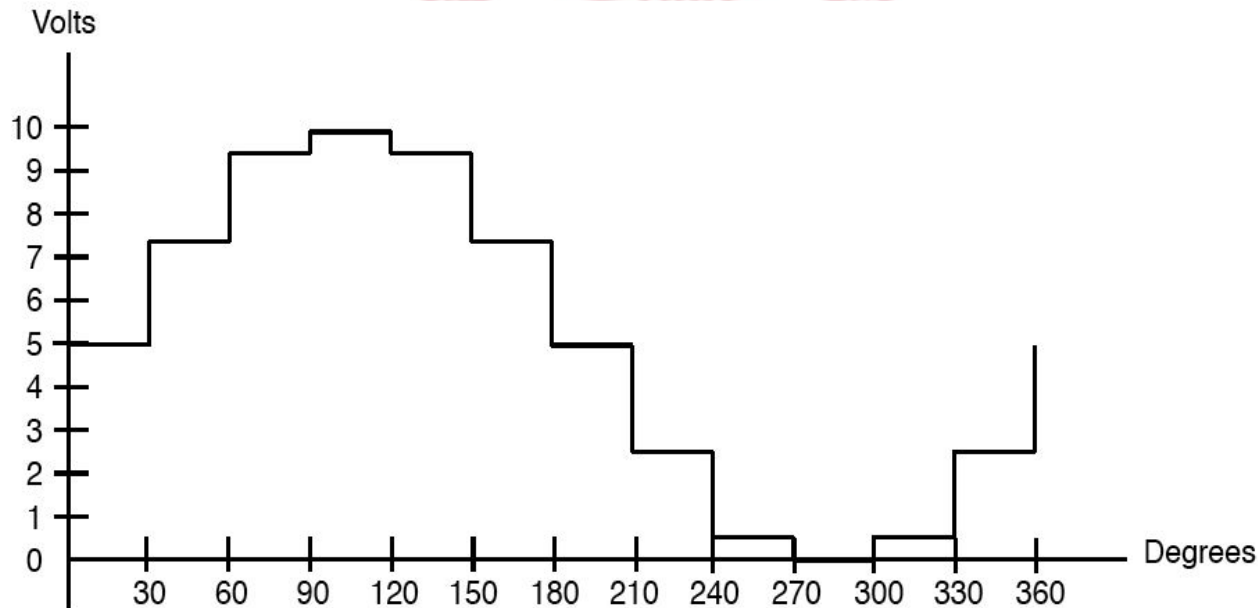
$$V_{out} * 25.6 = \text{Decimal value to be sent to DAC}$$

Calculations for each value of angle

Angle	Sin θ	Vout	Decimal
0	0	5	128
30	0.5	7.5	192
60	0.866	9.33	238
90	1.0	10	255
120	0.866	9.33	238
150	0.5	7.5	192
180	0	5	128
210	-0.5	2.5	64
240	-0.866	0.699	17
270	-1	0	0
300	-0.866	0.699	17
330	-0.5	2.5	64
360	0	5	128

Program to generate Sine wave

To generate a sine wave, table gives the magnitude of the sine angles between 0 and 360 degree angles. Vout of DAC for various sine degrees is calculated as: $V_{out} = 2.5 \text{ V} + (2.5 \text{ V} \times \sin \theta)$



Program to generate Sine wave

```
#include <reg51.h>
sfr DACDATA = P1;
void main()
{
    unsigned char WAVEVALUE[12] = {128,192,238,255,
                                    238,192,128,64,
                                    17,0,17,64};

    unsigned char x;
    while(1)
    {
        for(x=0;x<12;x++)
        {
            DACDATA = WAVEVALUE[x];
        }
    }
}
```

Temperature sensor



Program to read temperature from ADC and convert it to decimal

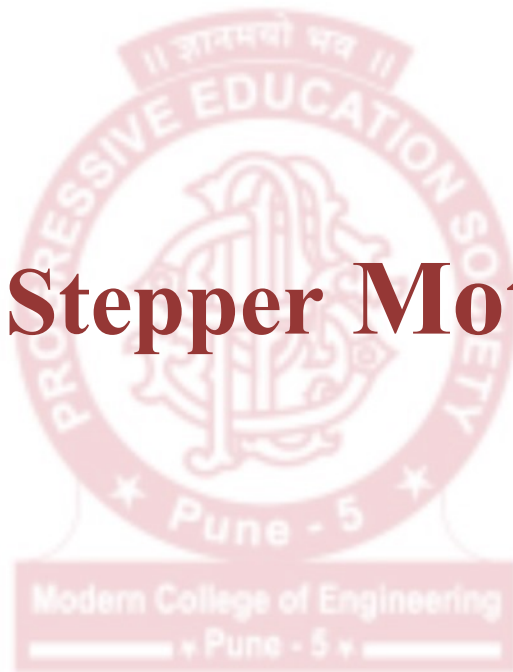
```
//C code to read temp from ADC0848, convert it to
//decimal, and put it on P0 with some delay
#include <reg51.h>
sbit RD = P2^5;
sbit WR = P2^6;
sbit INTR = P2^7;
sfr MYDATA = P1;      //P1 connected to D0-D7 of '848
void ConvertAndDisplay(unsigned char value);
void MSDelay(unsigned int value);
void main()
{
    MYDATA = 0xFF;      //make P1 and input
    INTR = 1;           //make INTR and input
    RD = 1;             //set RD high
    WR = 1;             //set WR high
    while(1)
    {
        WR = 0;        //send WR pulse
        WR = 1;
        while(INTR == 1); //wait for EOC
        RD = 0;         //send RD pulse
        value = MYDATA; //read value from ADC0848
        ConvertAndDisplay(value);
        RD = 1;
    }
}
```

```
void ConvertAndDisplay(unsigned char value)
{
    unsigned char x,d1,d2,d3;
    x=value/10;
    d1=value%10;
    d2=x%10
    d3=x/10
    P0=d1;                      //LSByte

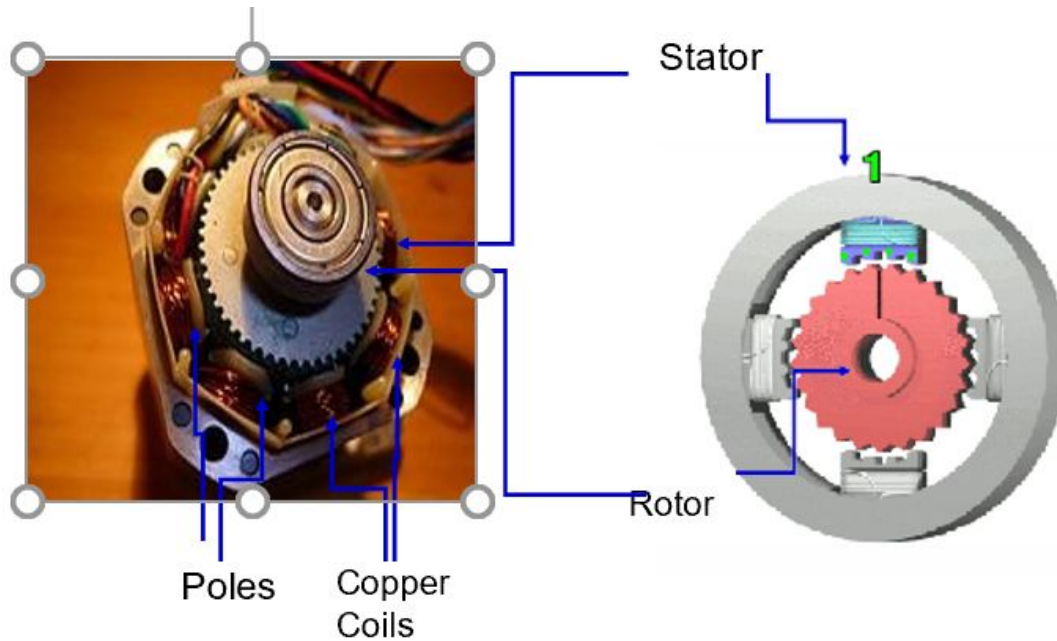
    MSDelay(250);
    P0=d2;
    MSDelay(250);
    P0=d3;                      //MSByte
    MSDelay(250);
}

void MSDelay(unsigned int value)
{
    unsigned char x,y;
    for(x=0;x<value;x++)
        for(y=0;y<1275;y++);
}
```

Stepper Motor



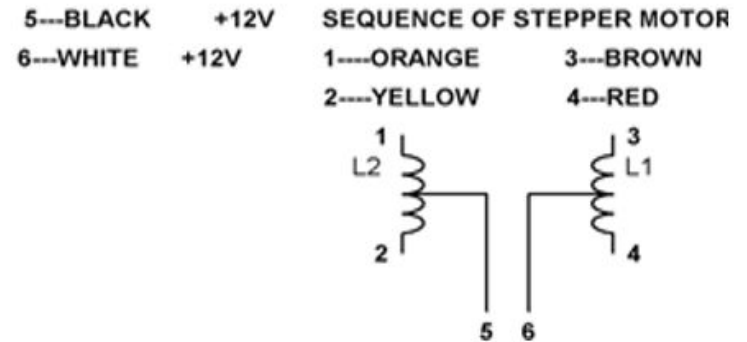
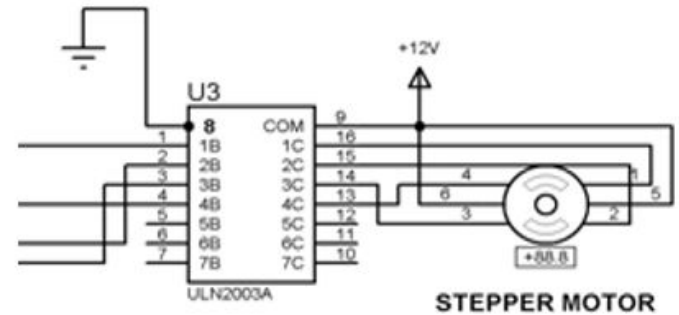
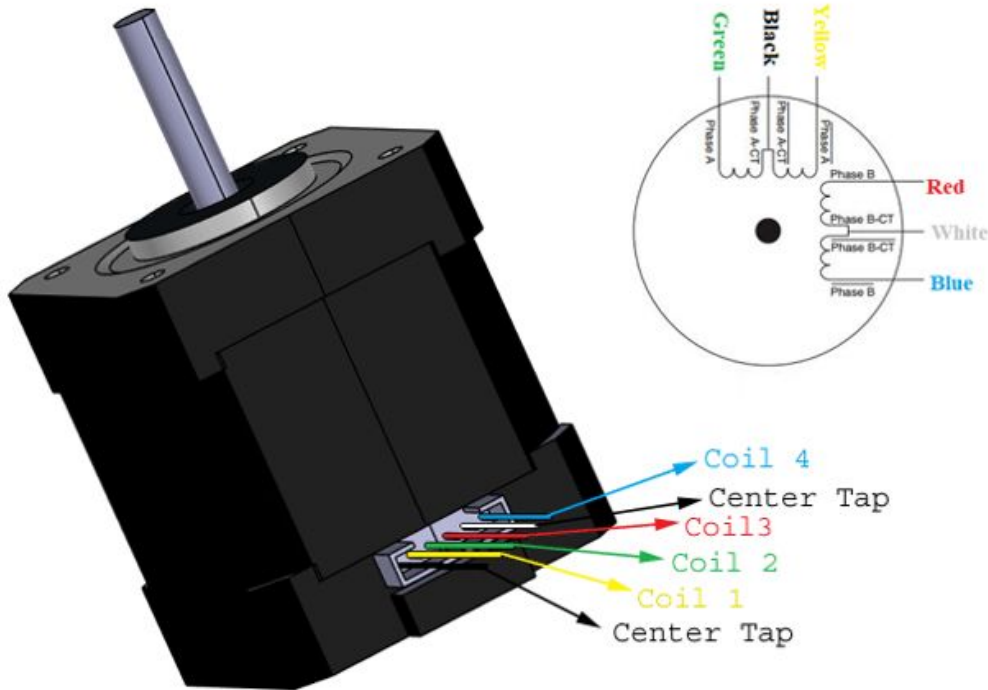
Stepper Motor



- . Rated Voltage: 12V DC
- . Current: 1.2A at 4V
- . Step Angle: 1.8 deg.
- . No. of Phases: 4
- . Motor Length: 1.54 inches
- . 4-wire, 8 inch lead
- . 200 steps per revolution, 1.8 degrees
- . Operating Temperature: -10 to 40 °C
- . Unipolar Holding Torque: 22.2 Nm

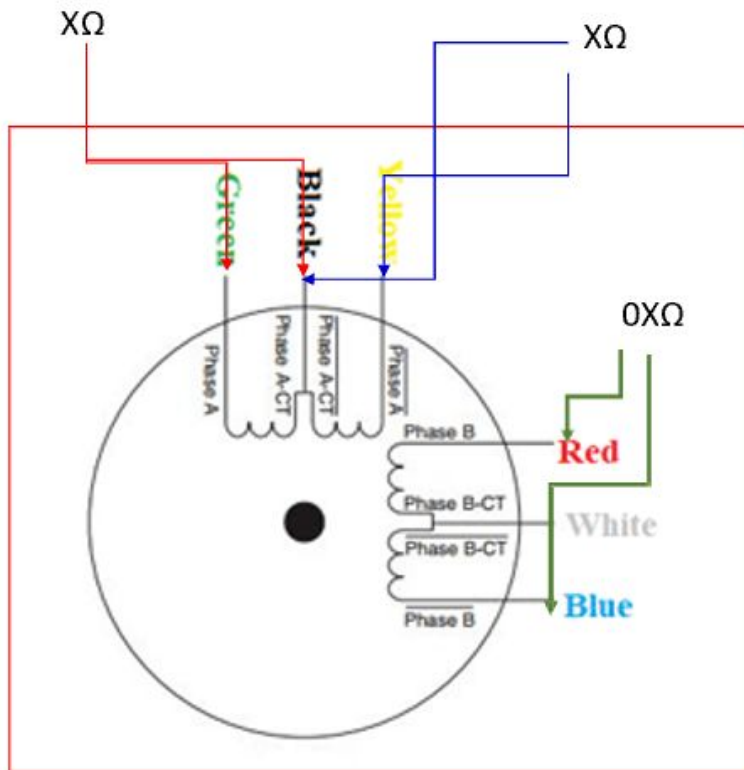
Courtesy: Google images

Stepper Motor



Courtesy: Google images

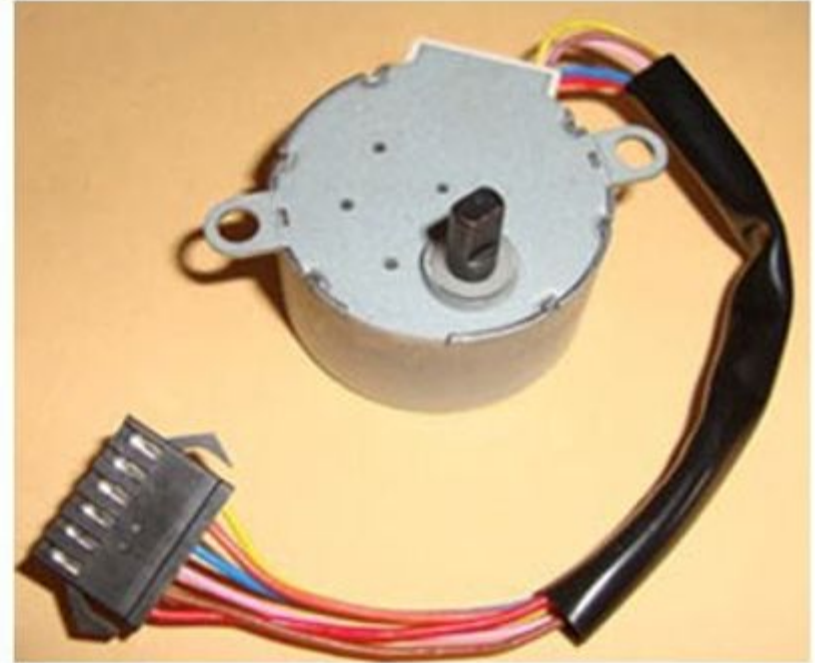
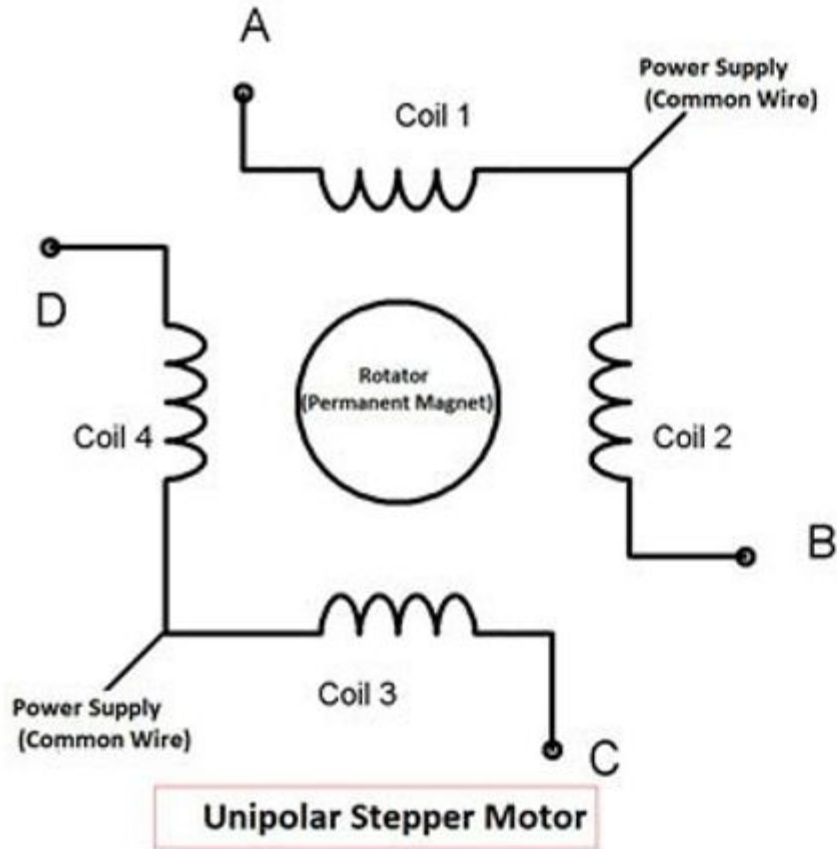
Stepper Motor



- We need to find out four coil wires and two common wires very carefully otherwise motor will not rotate.
- You can find it out by measuring resistance using multimeter, multimeter won't show any readings between the wires of two phases.

Courtesy: Google images

Stepper Motor



Courtesy: Google images

Interfacing of Stepper motor with 8051

A **motor** is one which translates electrical pulses into mechanical motion. Types of motor are:

Stepper Motor

DC Motor

AC Motor

A **stepper motor** is a special type of electric motor that moves in increments, or steps, rather than turning smoothly as a conventional motor does.

Typical increments are 0.9 or 1.8 degrees, with 400 or 200 increments thus representing a full circle.

The speed of the motor is determined by the time delay between each incremental movement.

Two types of stepper motor are:

Permanent Magnet (PM), Variable Reluctance (VR)

Construction

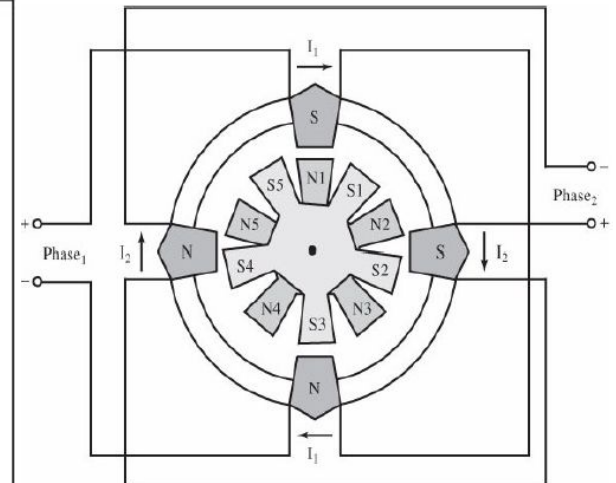
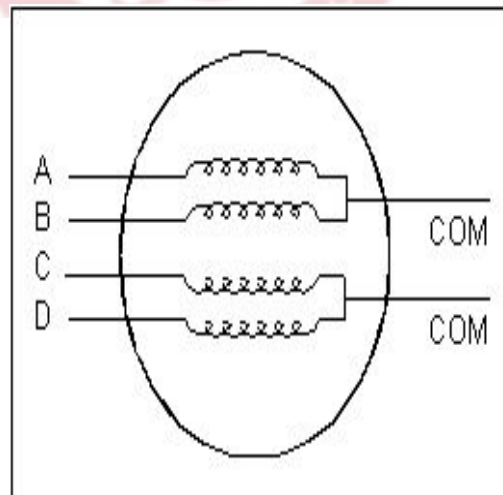
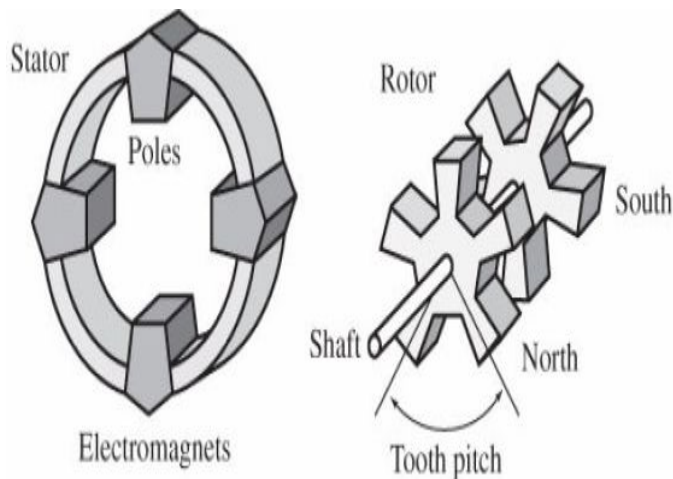
Permanent Magnet Stepper motor

This type of motor has rotor and stator

Permanent Magnet Rotor is also called the Shaft
Stator

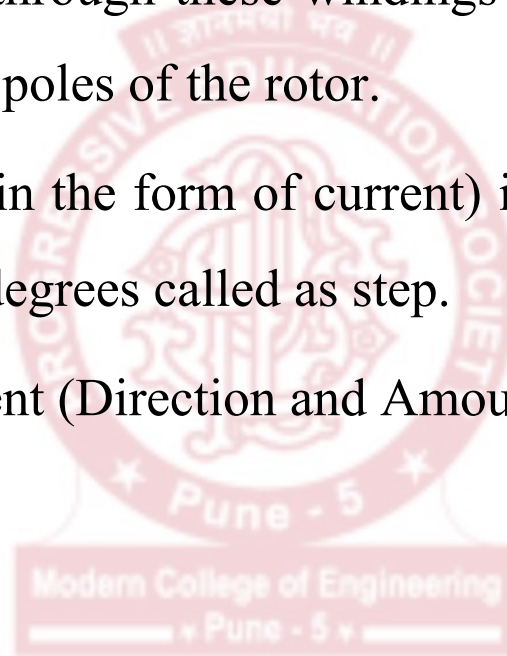
Surrounds the Shaft

Usually Four Stator Windings Paired with Center-Tapped Common
known as Four-Phase or Unipolar Stepper Motor.



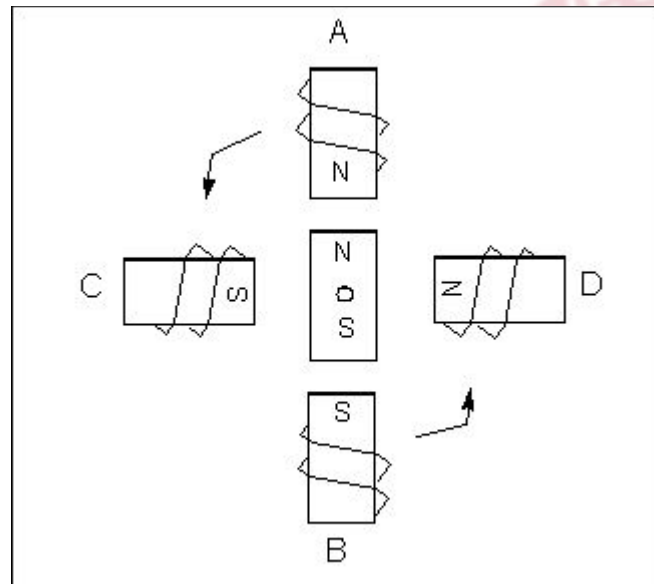
Working Principle

- In permanent Magnet motor the rotor is made up of permanent magnet.
- When current flows through these windings it will produce magnetic flux which will attract the poles of the rotor.
- Each Time a Pulse (in the form of current) is received rotor turns through specified number of degrees called as step.
- Can Control Movement (Direction and Amount) Easily

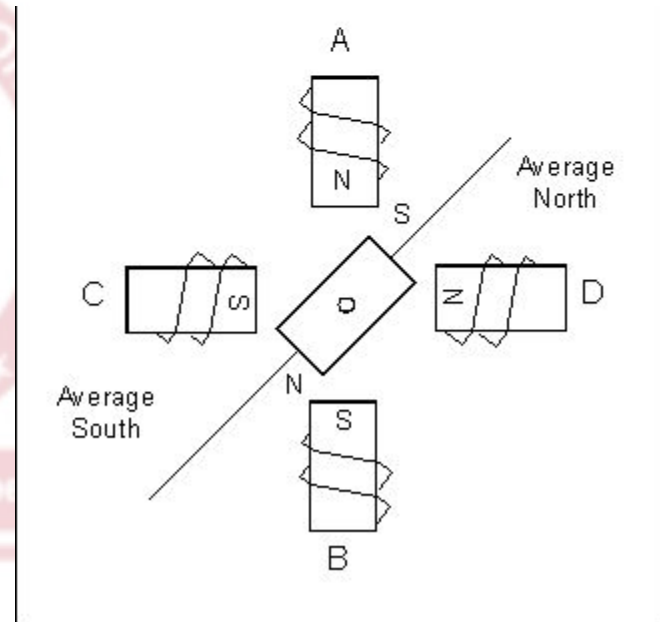


Moving the Rotor

Unstable



Stable



Rotor will ALWAYS seek a *stable* position.

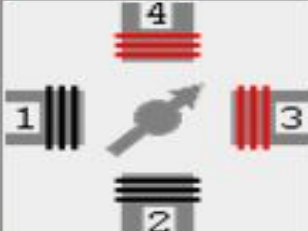
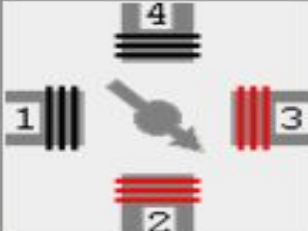
Wave Drive Mode

Single-Coil Excitation - Each successive coil is energized in turn

Step	Coil 4	Coil 3	Coil 2	Coil 1	
a.1	on	off	off	off	
a.2	off	on	off	off	
a.3	off	off	on	off	
a.4	off	off	off	on	

Full Drive Mode

Two-Coil Excitation - Each successive pair of adjacent coils is energized in turn

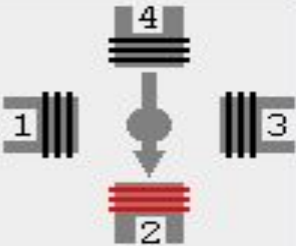
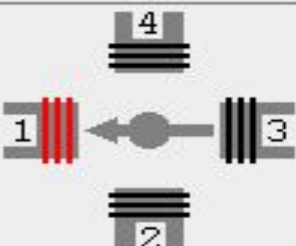
Step	Coil 4	Coil 3	Coil 2	Coil 1	
b.1	on	on	off	off	
b.2	off	on	on	off	
b.3	off	off	on	on	
b.4	on	off	off	on	

Half Drive Mode

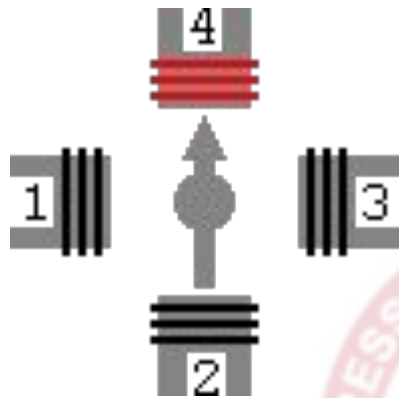
Interleaving the two sequences will cause the motor to half-step
 8 step sequence = normal 4 step + wave drive 4 step.

Step	Coil 4	Coil 3	Coil 2	Coil 1	
a.1	on ON	off	off	off	
b.1	on ON	on ON	off	off ON	
a.2	off	on ON	off	off	
b.2	off	on ON	on ON	off	

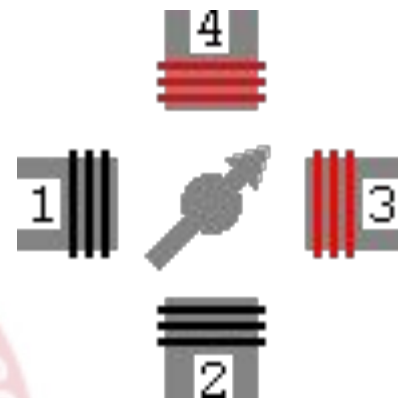
Half Drive Mode

Step	Coil 4	Coil 3	Coil 2	Coil 1	
a.3	off	off	on	off	
b.3	off	off	on	on	
a.4	off	off	off	on	
b.4	on	off	off	on	

All 3 Modes of operation

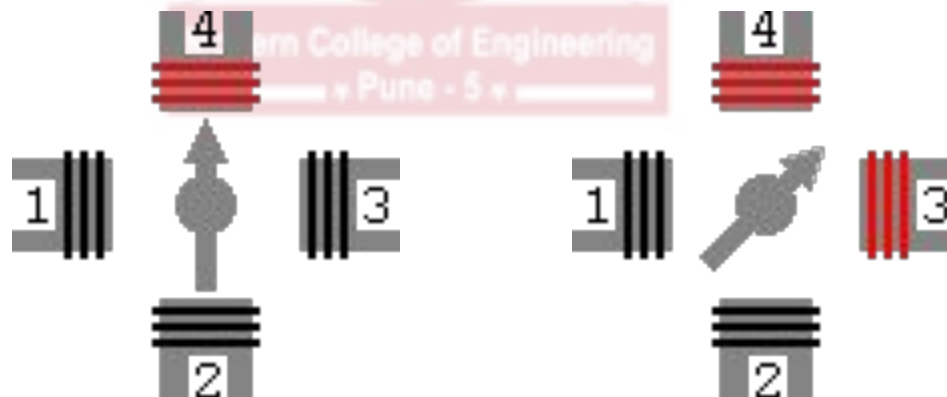


Single-Coil Excitation



Two-Coil Excitation

Interleaved Single- and Two-Coil Excitation



Description of 3 Modes of operation

<u>Sequence</u>	<u>Name</u>	<u>Description</u>
0001 0010 0100 1000	Wave Drive, One-Phase	Consumes the least power. Only one phase is energized at a time. Assures positional accuracy regardless of any winding imbalance in the motor.
0011 0110 1100 1001	Hi-Torque, Two-Phase	Hi Torque - This sequence energizes two adjacent phases, which offers an improved torque-speed product and greater holding torque.
0001 0011 0010 0110 0100 1100 1000 1001	Half-Step	Half Step - Effectively doubles the stepping resolution of the motor, but the torque is not uniform for each step. (Since we are effectively switching between Wave Drive and Hi-Torque with each step, torque alternates each step.) This sequence reduces motor resonance which can sometimes cause a motor to stall at a particular resonant frequency. Note that this sequence is 8 steps.

How Far Does It Move?

Step Angle:

Arc Through Which Motor Turns With ONE Step Change of the Windings

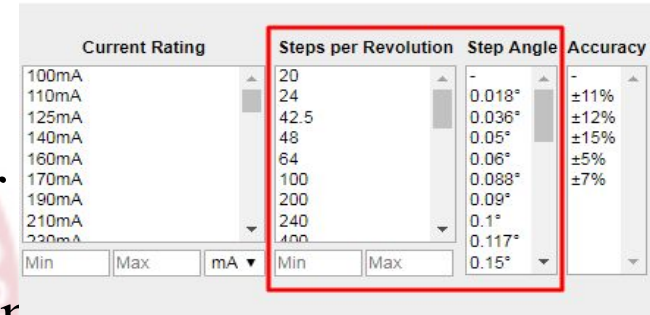
Varies With Model of Stepper Motor

(Depending on the number of teeth on stator and rotor)

Normally in Degrees

Step angle = $360/\text{No. of Steps per Revolution}$

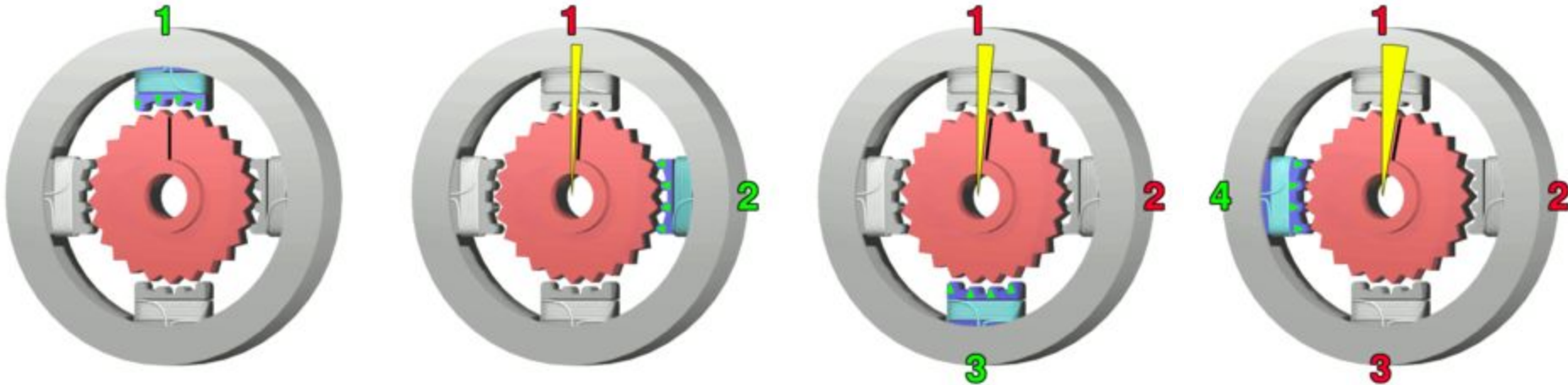
Commonly available no. of steps per revolution are 500, 200, 180, 144, 72, 48, 24



Current Rating	Steps per Revolution	Step Angle	Accuracy
100mA	20	-	-
110mA	24	0.018°	±11%
125mA	42.5	0.036°	±12%
140mA	48	0.05°	±15%
160mA	64	0.06°	±5%
170mA	100	0.036°	±7%
190mA	200	0.09°	-
210mA	240	0.1°	-
230mA	400	0.117°	-
Min	Max	0.15°	-

How Fast?

$$rpm = \frac{60 \times \text{Steps per Second}}{\text{Steps per Revolution}}$$

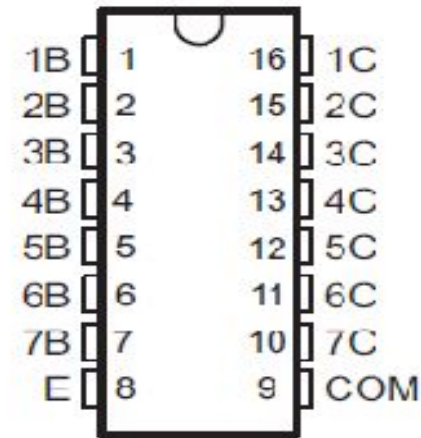
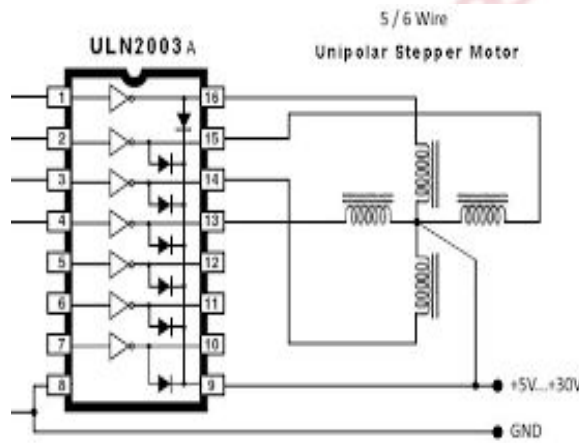


Interfacing of Stepper Motor with 8051

- Port P1 of 8051 for connecting the stepper motor.
- A Stepper motor consumes a current of 0.1 – 1 A during step rotation with the load.
- An 8051 produces a maximum current of 0.045A through the ports.
- Hence, we cannot directly interface stepper motors with microcontrollers
- To boost the current, ULN2003 is used which is a high voltage, high current Darlington transistor array.
- ULN2003 is an IC containing pairs of Darlington configuration of transistors.
- The Darlington transistor is made up of two PNP or NPN BJTs by connecting together where the emitter of the PNP transistor is connected to the base of the other PNP transistor to create a sensitive transistor with high current gain
- ULN2003 takes low power input and outputs a signal with greater power

ULN2003

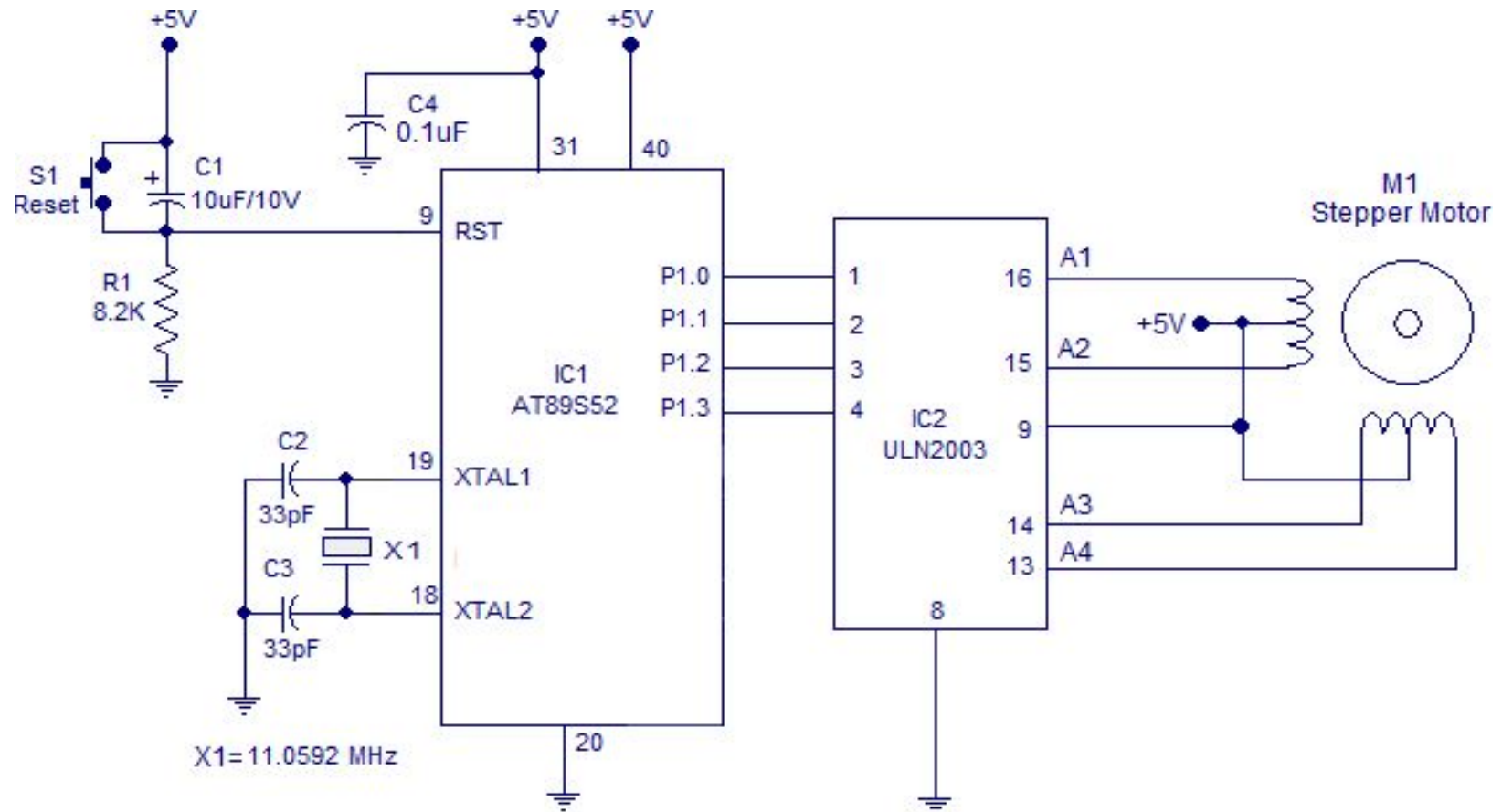
- ULN2003A is the array of seven NPN Darlington transistor pairs. Darlington pair is constructed by connecting two bipolar transistors to achieve high current amplification. In ULN2003A, 7 pins are input pins and 7 pins are output pins, two pins are for Vcc (power supply) and Ground. Here we are using four input and four output pins. We can also use L293D IC in place of ULN2003A for current amplification.



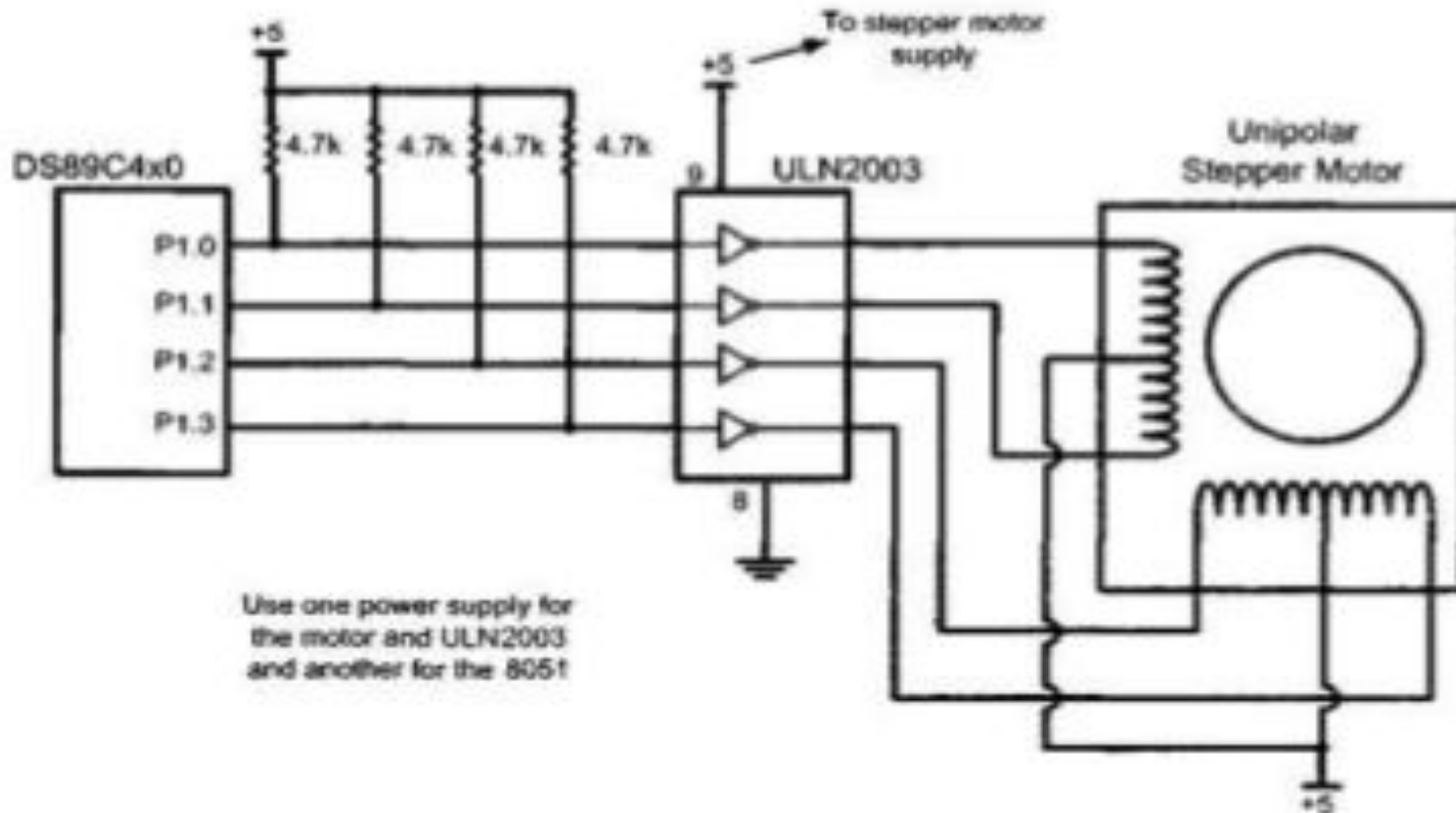
Features :

- 500-mA-Rated Collector Current (Single Output)
- High-Voltage Outputs: 50 V
- Output Clamp Diodes
- Inputs Compatible With Various Types of Logic
- Can be directly controlled by logic level 0 or 1

Interfacing of Stepper Motor with 8051



Interfacing of Stepper Motor with 8051 using opto-isolator



Parameters of stepper motors

- Step Angle – The step angle is the angle in which the rotor moves when one pulse is applied as an input of the stator. This parameter is used to determine the positioning of a rotor.
- Steps per Revolution – This is the number of step angles required for a complete revolution. So the formula is $(360^\circ / \text{Step Angle})$.
- Steps per Second – This parameter is used to measure a number of steps covered in each second.
- RPM – The RPM is the Revolution Per Minute. It measures the frequency of rotation. By this parameter, we can measure the number of rotations in one minute.
- The relation between RPM, steps per revolution, and steps per second is like below:
$$\text{Steps per Second} = \text{rpm} \times \text{steps per revolution} / 60$$

Program to rotate a motor 45° forward and 30° reverse

n Steps = 360 degrees 1 Step = 'x' degrees

$x = 360 / n \text{ degree}$

n Steps = 360 (1 revolution) / step size

= 360/45

= 8

n Steps = 360 (1 revolution) / step size

= 360/30

= 12

```
ORG 00H
MOV R0, #8      ;for 45 degree
MOV A, #66H;
ACALL DELAY
AGAIN: RRC A    ;Forward move
ACALL DELAY
DJNZ R0, AGAIN
```

```
ORG 00H
MOV R0, #12      ;for 30 degree
MOV A, #66H;
ACALL DELAY
AGAIN: RLC A    ;Reverse move
ACALL DELAY
DJNZ R0, AGAIN
```

```
DELAY: MOV R1, #200
BACK: MOV R2, #250
HERE: DJNZ R2, HERE
      DJNZ R1, BACK
      RET
```

DC Motor	Stepper Motor
- It has a frame that remains motionless and an armature that moves (like EM relay and solenoids) - Armature moves in a circular manner (shaft rotation). -Brushes positioned between the frame and armature are used to alternate the current direction through the coil so that a DC current generates a continuous rotation of the shaft. - When the current is removed, the magnetic force stops and the shaft is free to rotate. - In a PWM DC motor, the computer activates the coil with a current of fixed magnitude but varies the duty cycle to control the motor speed.	- Both the position and velocity of a stepper motor can be controlled using a microcontroller in open-loop fashion. - Stepper motors may not require feedback sensor, so cost is less as compared to a DC motor. - Used in hard disk drives, floppy disk drives, printers and can also be used as shaft encoders for measuring both position and speed.

Motion Detectors

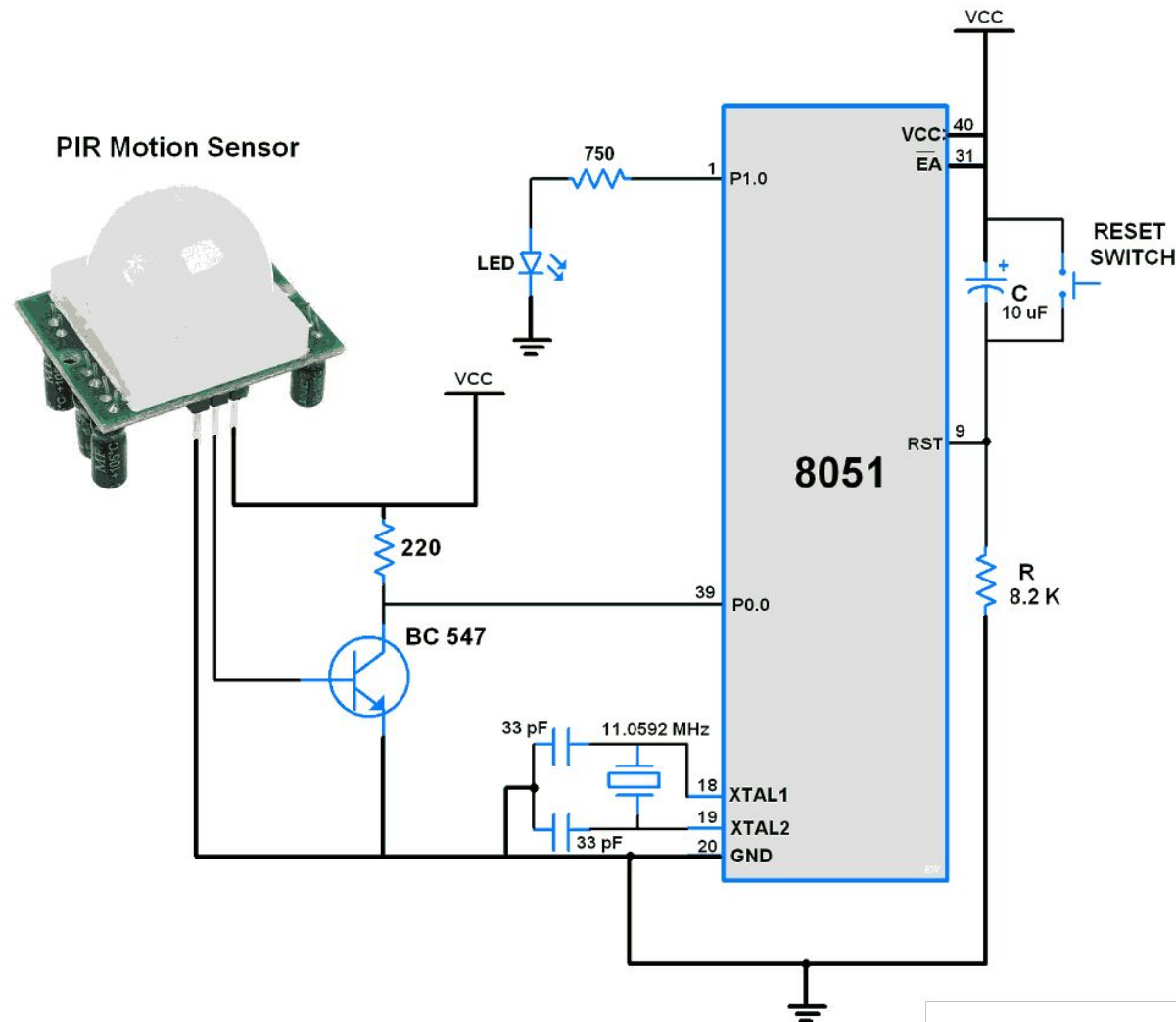
Motion Sensor

- A motion detector is a device that detects moving objects particularly people.
- Such device is often integrated as a component of a system that automatically performs a task or alerts.
- Domestic and commercial applications
 - Activating automatic door openers
 - Smart lighting systems
 - Burglar alarm
 - A security camera

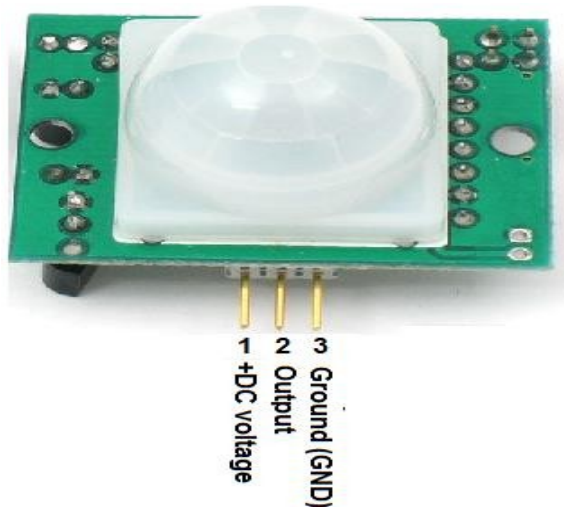
Motion Sensor

- PIR sensor detects infrared heat radiations.
- It can be used to detect the presence of living objects that emit infrared heat radiation.
- The PIR sensor is split into two slots.
- Whenever a stationary object is in front of the sensor, the two slots receive the same amount of radiation and the output is zero.
- Whenever a moving object is in front of the sensor, one of the slots receives more radiation than the other slot.
- This makes the output swing high or low.
- This change in output voltage is the result of the detection of motion.

Interfacing of PIR Motion sensor with 8051 Microcontroller



Interfacing of PIR Motion sensor with 8051 Microcontroller



```
#include <reg51.h>
sbit Motion_detection=P0^0;      /* Read PIR sensor's data on this pin */
sbit LED=P1^0;                   /* Connect LED to the PORT1.0 pin */
void MSdelay(unsigned int val);

void main(void)
{
    P1=0;                        /* Initially LED turned OFF*/
    MSdelay(3000);               /* Power-on delay for PIR */
    while(1)
    {
        if(Motion_detection==1) /* Check for human motion */
            LED = 0;             /* LED turn OFF for No motion */
        else
            LED = 1;             /* LED turn ON if motion is detected */
    }
}

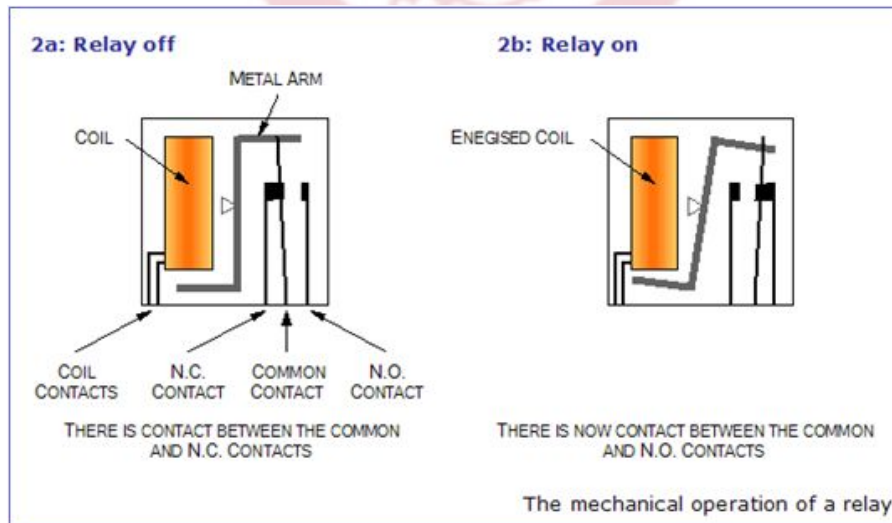
void MSdelay(unsigned int val)
{
    unsigned int i,j;
    for(i=0;i<=val;i++)
        for(j=0;j<112;j++);     /* Delay of 1 ms for 11.0592MHz Frequency */
}
```



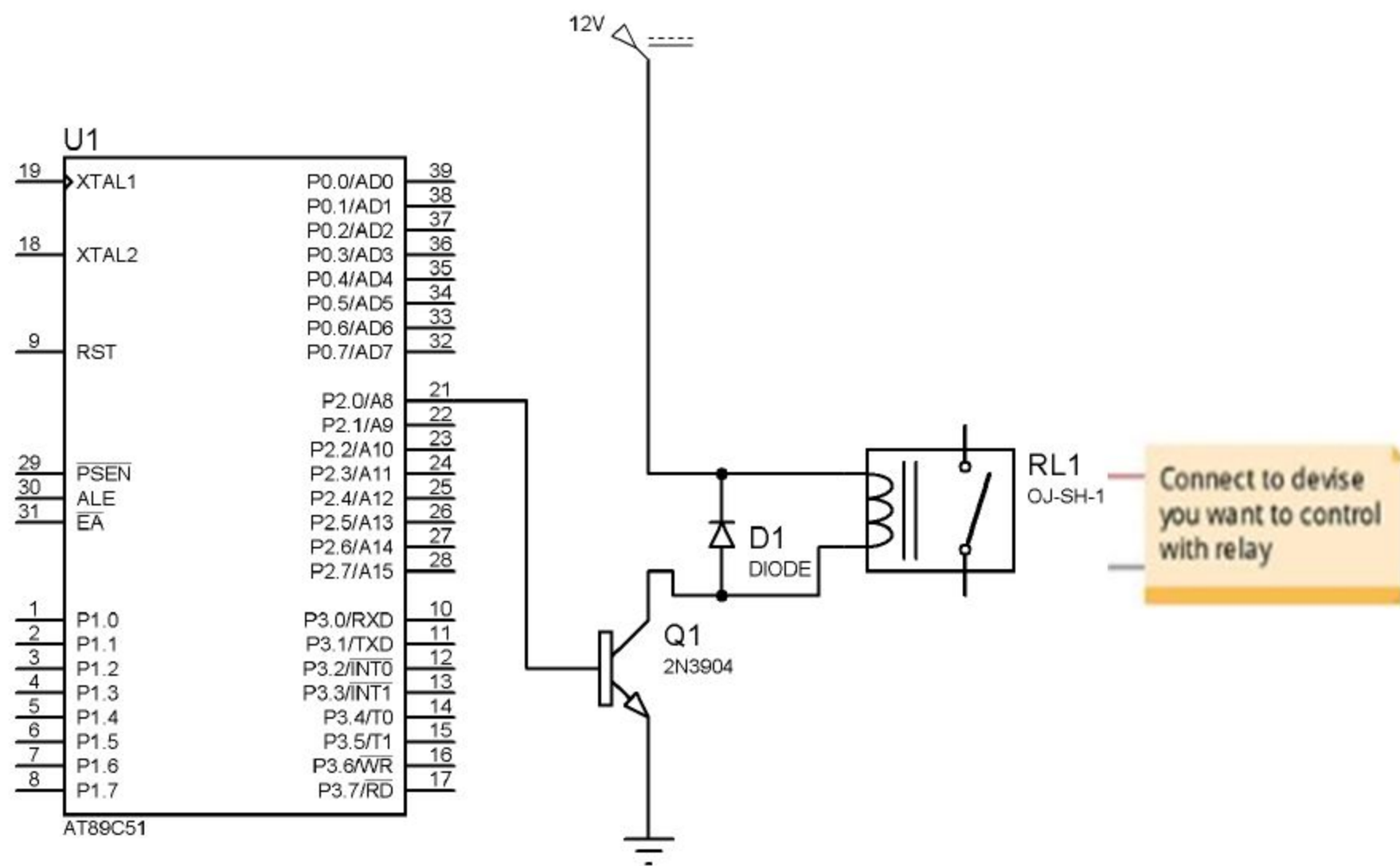
Relay

Relay

- Electromechanical Relay
 - Made up of coil and armature. Thus its life is small.
- Solid state relay
 - Made up of semiconductor material
 - Operate at extremely low current and small packaging, thus ideal for all microprocessors and controllers
 - Widely used in controlling lamps, fans, and all power applications.

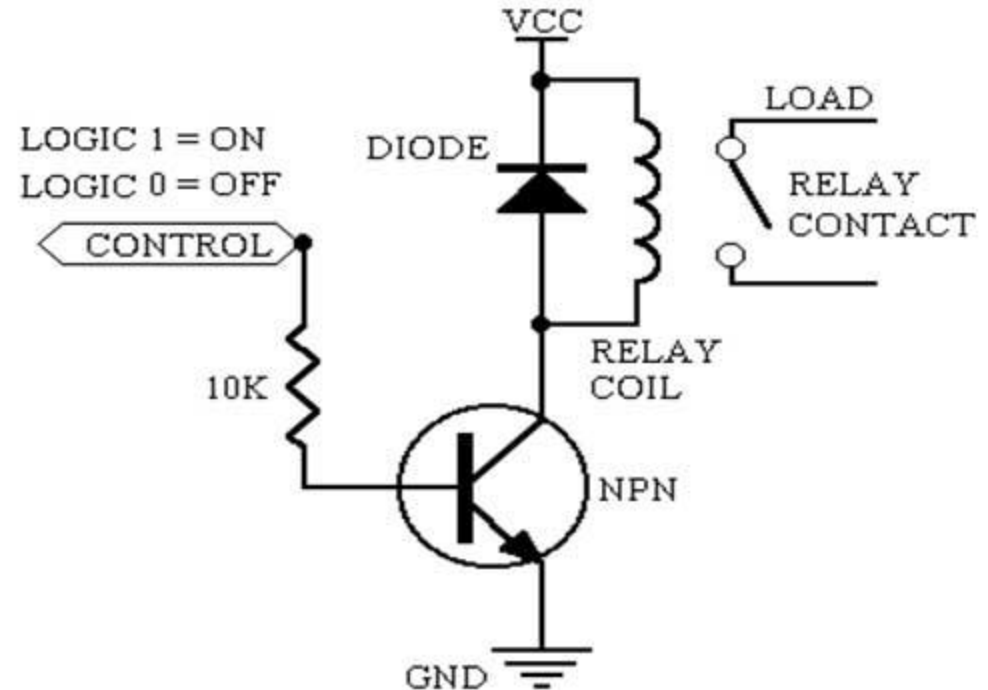


Interfacing diagram of (electromechanical relay) with 8051



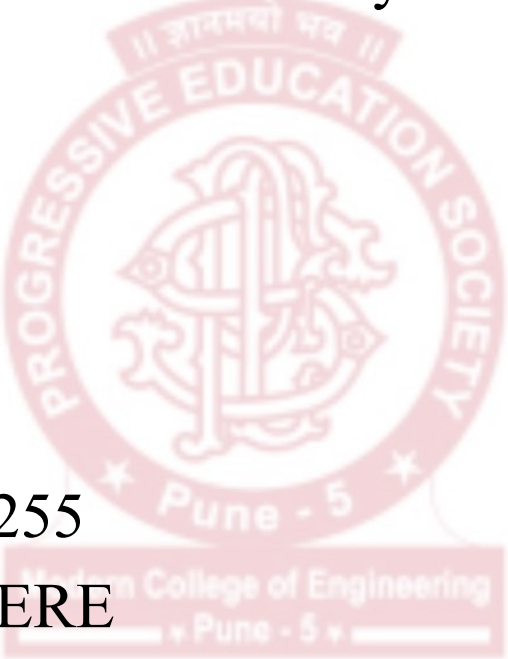
Interfacing diagram of (electromechanical relay) with 8051

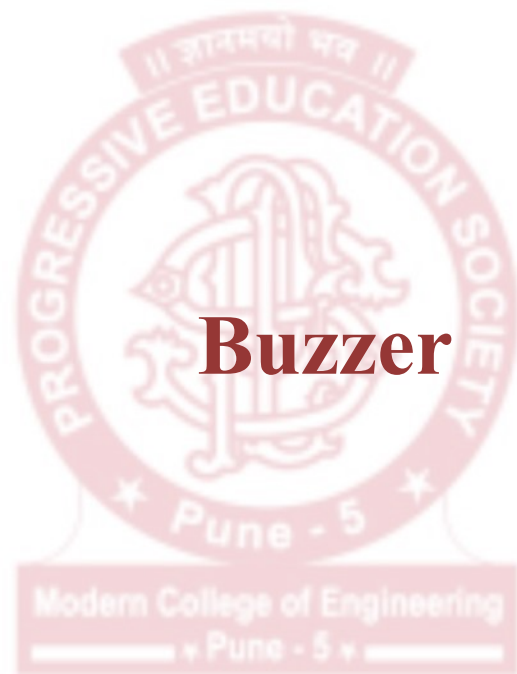
- The freewheeling diode freewheels the energy stored in the coil of the relay when the transistor turns off from its on state.
- If there is not a diode, then the current in the coil abruptly stops flowing, which causes a high voltage spike, that can cause damage to the transistor driving the relay



Program

```
ORG 0000H
SETB P2.0      ; terminal to relay to appliance
ACALL DELAY
CLR P2.0
ACALL DELAY
DELAY:
    MOV R1, #255
HERE1:MOV R2, #255
HERE: DJNZ R2, HERE
      DJNZ R1, HERE1
      RET
```





Buzzer

Buzzer



Mini Piezo Buzzer
pcboard.ca · In stock



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indiamart.com



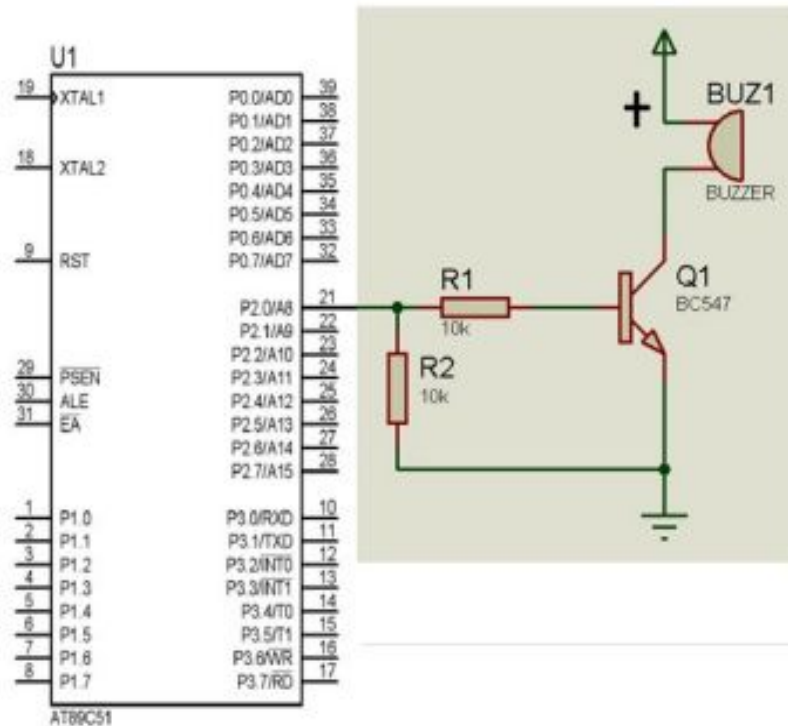
Large Size Piezoelectric Buzze...
roboelements.com · In stock



Continuous Sound Beeper Pie...
aliexpress.com · In stock



Interfacing of Buzzer with 8051 and program



```

ORG 0000h
CLR P2.0                                ;Initially make relay off

AGAIN:  ACALL DELAY

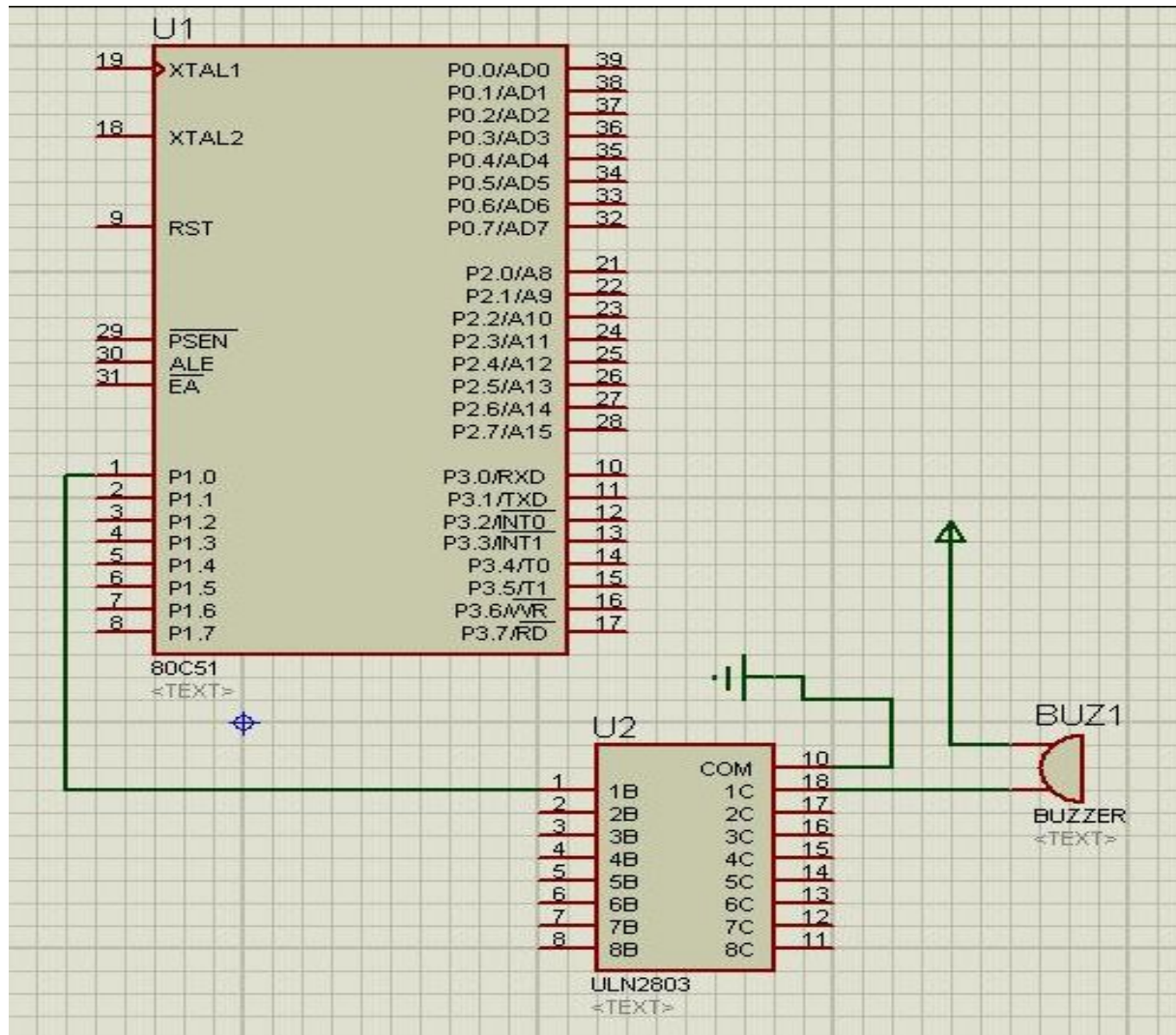
CPL P2.0                                ; Make it on-off

SJMP AGAIN

ORG 0000h
AGAIN:  CLR P2.0                        ;Initially make
relay off

        ACALL DELAY
        SETB P2.0                      ; Make it ON
        ACALL DELAY
        ACALL DELAY
        SJMP AGAIN
    
```

Interfacing of Buzzer with 8051 using IC

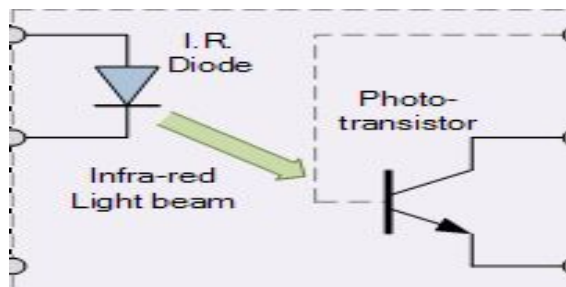




Opto-isolator

Opto-isolator

- Optocouplers can be used to isolate low-power circuits from higher power circuits and to remove electrical noise from signals.
- The basic design of an optocoupler, also known as an Opto-isolator, consists of an LED that produces infra-red light and a semiconductor photo-sensitive device that is used to detect the emitted infra-red beam.
- Both the LED and photo-sensitive device are enclosed in a light-tight body or package with metal legs for the electrical connections as shown.
- Opto-isolator is used to protect a low voltage device (8051) from high voltage device (bulb). The purpose of an optocoupler is to transfer signals from one circuit to another yet keep them separate so that there is no direct current flow.



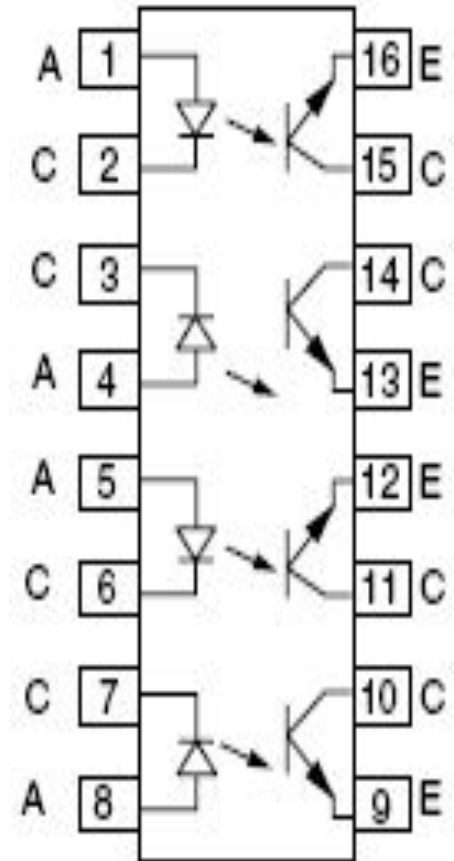
Opto-isolator

- An opto-isolator or optocoupler is used to isolate two parts of a system.
- An example is driving a motor. Motors can produce what is called back EMF, a high voltage spike produced by a sudden change of current as indicated in the $V = L di/dt$ formula
- An opto-isolator has an LED (light-emitting diode) transmitter and a photosensor receiver, separated from each other by a gap.
- When current flows through the diode, it transmits a signal light across the gap and the receiver produces the same signal with the same phase but a different current and amplitude.
- The gap between the transmitter and receiver of opto-isolators prevents the electrical current surge from reaching the system.

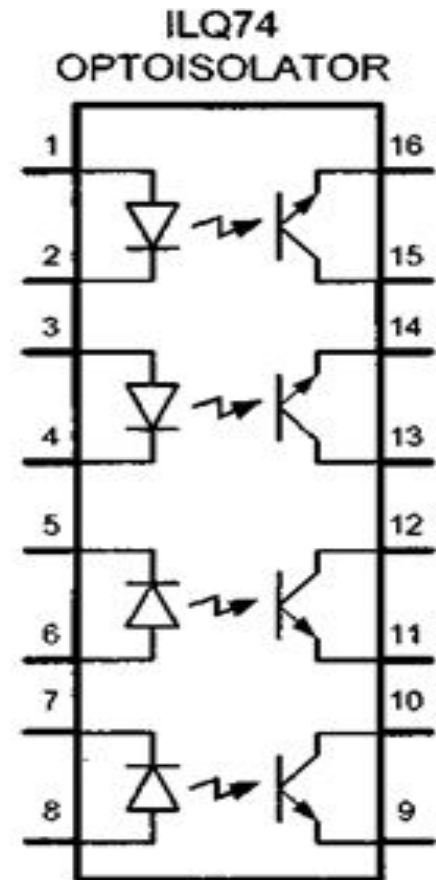
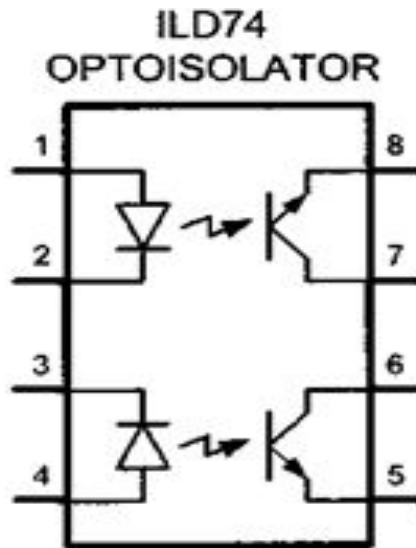
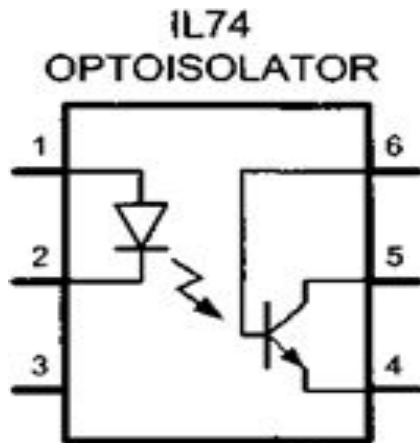
Working

- When an electrical current is applied to the LED, infrared light is produced and passes through the material inside the opto-isolator.
- The beam travels across a transparent gap and is picked up by the receiver, which converts the modulated light or IR back into an electrical signal.
- In the absence of light, the input and output circuits are electrically isolated from each other.

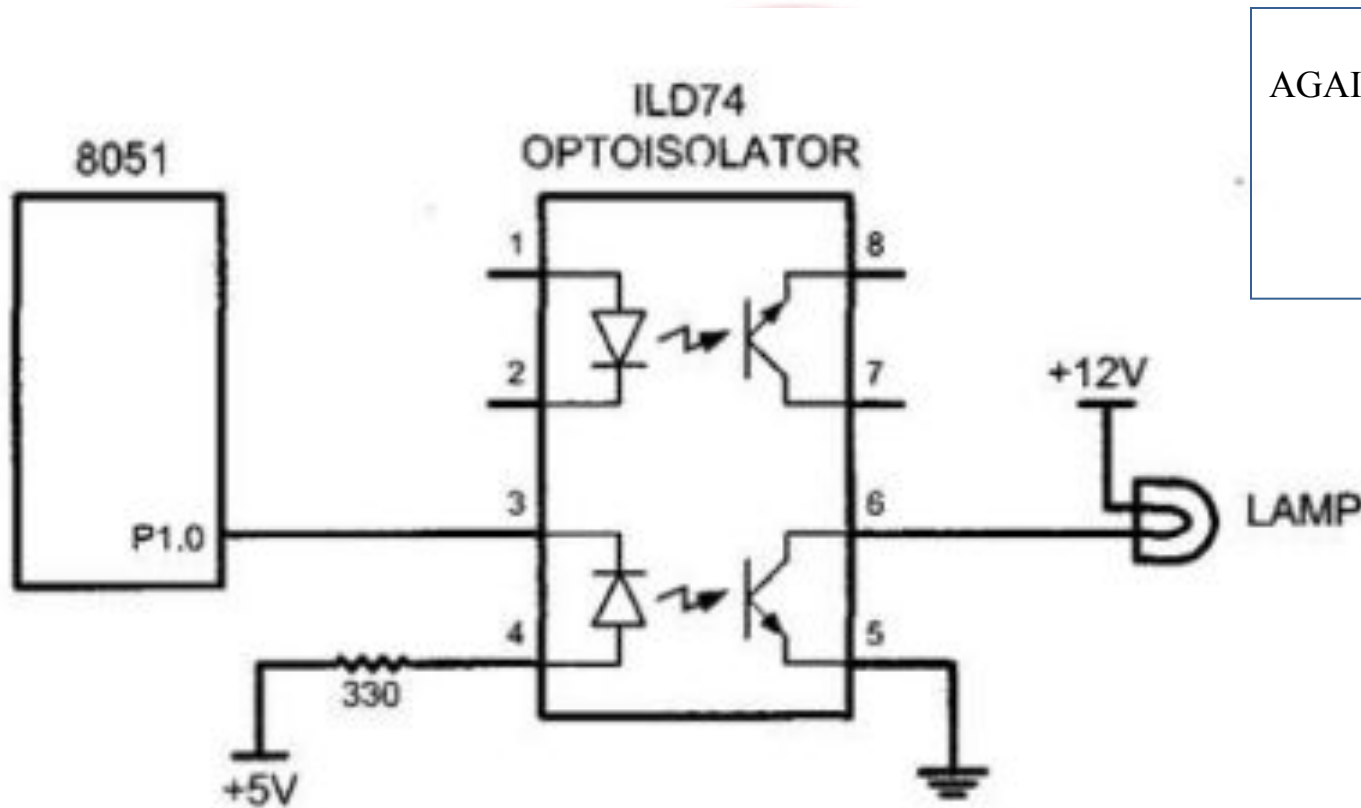
Pin-out



Opto-isolator



Interfacing of Opto-isolator with 8051 to lit lamp



```
ORG 0000H  
AGAIN : CLR P1.0  
        ACALL DELAY  
        SETB P1.0  
        ACALL DELAY  
        SJMP AGAIN
```

References:

- Textbooks prescribed in the university syllabus.
- As per the presentation slides received during Online Faculty Orientation Workshop under the aegis of BOS (E&TC), SPPU, Pune for syllabus discussions.

